



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Why does the effect vary?

Meta-regression, missing data and a glimpse of network meta-analysis

Damien Beillouin
CIRAD, UPR HortSys

SOUTIENS INSTITUTIONNELS





Learning objectives

By the end of this session, you will be able to:

- 1 Use **meta-regression** to test whether a moderator explains heterogeneity – and interpret it without over-claiming.
- 2 Tell **within-study** from **between-study** evidence for a moderator.
- 3 Handle **missing standard deviations** transparently.
- 4 Recognise when a **network meta-analysis** (or another advanced method) is needed.

The question left open at 14:00

Pests under intercropping: **-27%** on average...

...but in a new field anywhere from **-80%** to **+170%**.

Why?

Crop, pest species, climate, design, intercrop density... A **moderator** is a study or comparison characteristic that may explain part of this variation.

At 15:00 we already did it

The multilevel Egger test added precision as a moderator; the validity test added study quality. **Meta-regression = a meta-analysis with predictors.**

Meta-regression

From meta-analysis to meta-regression

This afternoon's multilevel model, **plus a predictor**:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + u_j + u_{ij} + \varepsilon_{ij}$$

- $\beta_1 x_{ij}$: the moderator (categorical or continuous)
- u_j, u_{ij} : what the moderator does **not** explain (residual heterogeneity, between and within studies)
- ε_{ij} : sampling error, known (v_{ij})

Three things to read

1. **Omnibus test** (F): does the moderator explain anything?
2. **Coefficients**: how much, in which direction?
3. **Pseudo- R^2** : how much of the heterogeneity is explained?

The same model in R

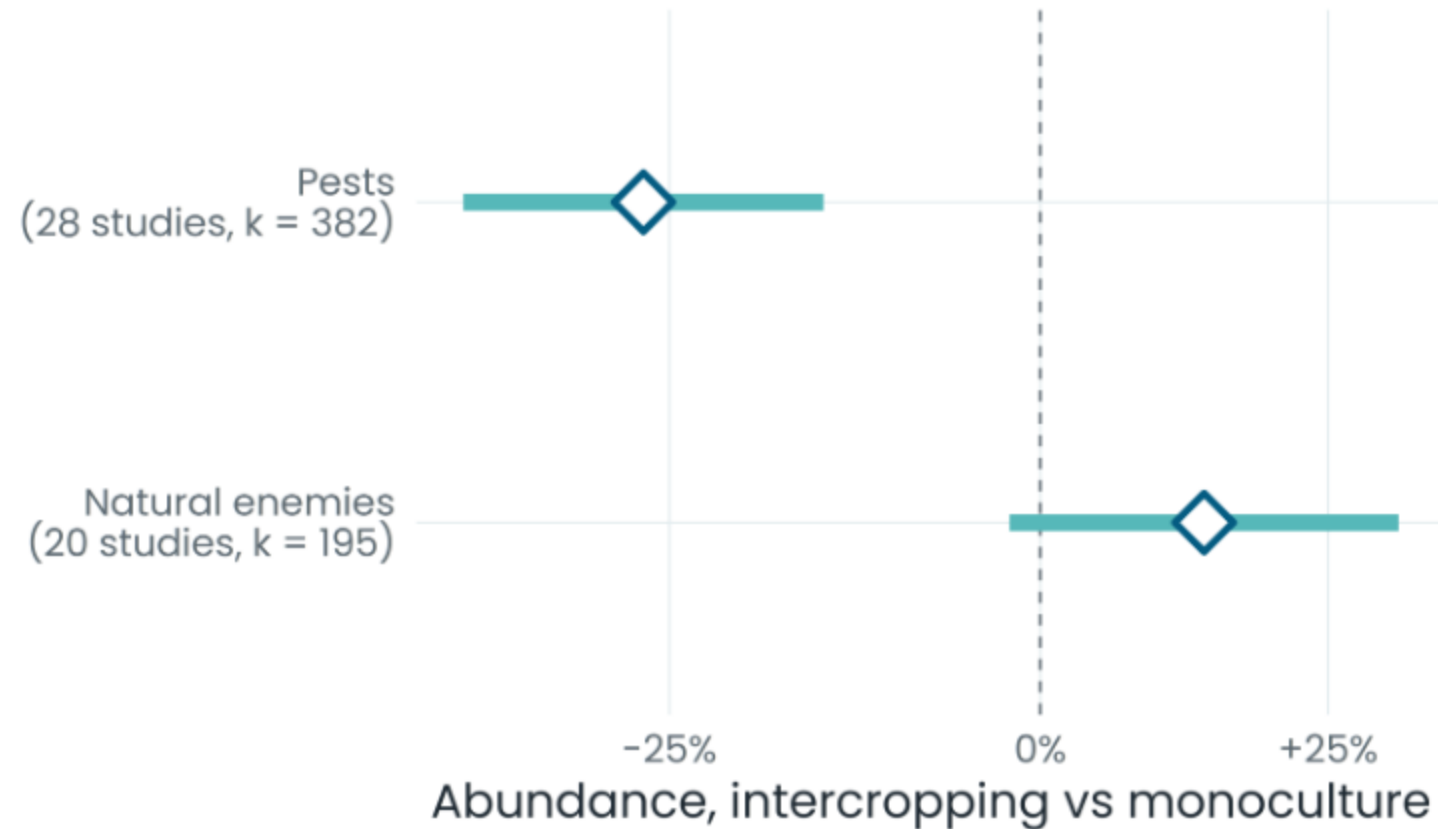
```
1 rma.mv(yi, vi,  
2       mods = ~ functional_group,  
3       random = ~ 1 | study_id / es_id,  
4       test = "t", dfs = "contain", data = dat)
```

In the equation	In the code	Meaning
$\beta_1 x_{ij}$	<code>mods = ~ functional_group</code>	the moderator
u_j	<code>study_id</code>	between-study heterogeneity
u_{ij}	<code>es_id</code>	within-study heterogeneity
ε_{ij}	<code>vi</code>	sampling error (known)

Pseudo- R^2 = share of the total heterogeneity removed by the moderator, computed on the **same rows**:

$$R^2 = 1 - \frac{\sum \sigma^2_{\text{with moderator}}}{\sum \sigma^2_{\text{without}}}$$

Pests vs natural enemies



- Difference: pests -35% relative to natural enemies
- Omnibus test, cluster-robust: $F = 20.4$ on 1 and 10 df, $p = 0.001$
- Pseudo- $R^2 \approx 10\%$ (15% of within-study, 2% of between-study variance)

A clear moderator... that explains only **~10%** of the heterogeneity. Most of the variation remains unexplained.

The groups, or the studies that measured them?

YOUR CALL

Studies on pests and studies on natural enemies may differ in crops, regions, designs... Can we trust the difference?

Between-study differences are **observational**: any study characteristic that travels with the moderator can **confound** it.

14 studies measured **both** groups. Split the moderator into its within- and between-study parts ($\text{mods} = \sim \text{pest_w} + \text{pest_m}$):
within studies: -36% · between studies: -28% ($p = 0.09$).

The within-study contrast — same fields, same year — is the stronger evidence.

Meta-regression: the rules of the game

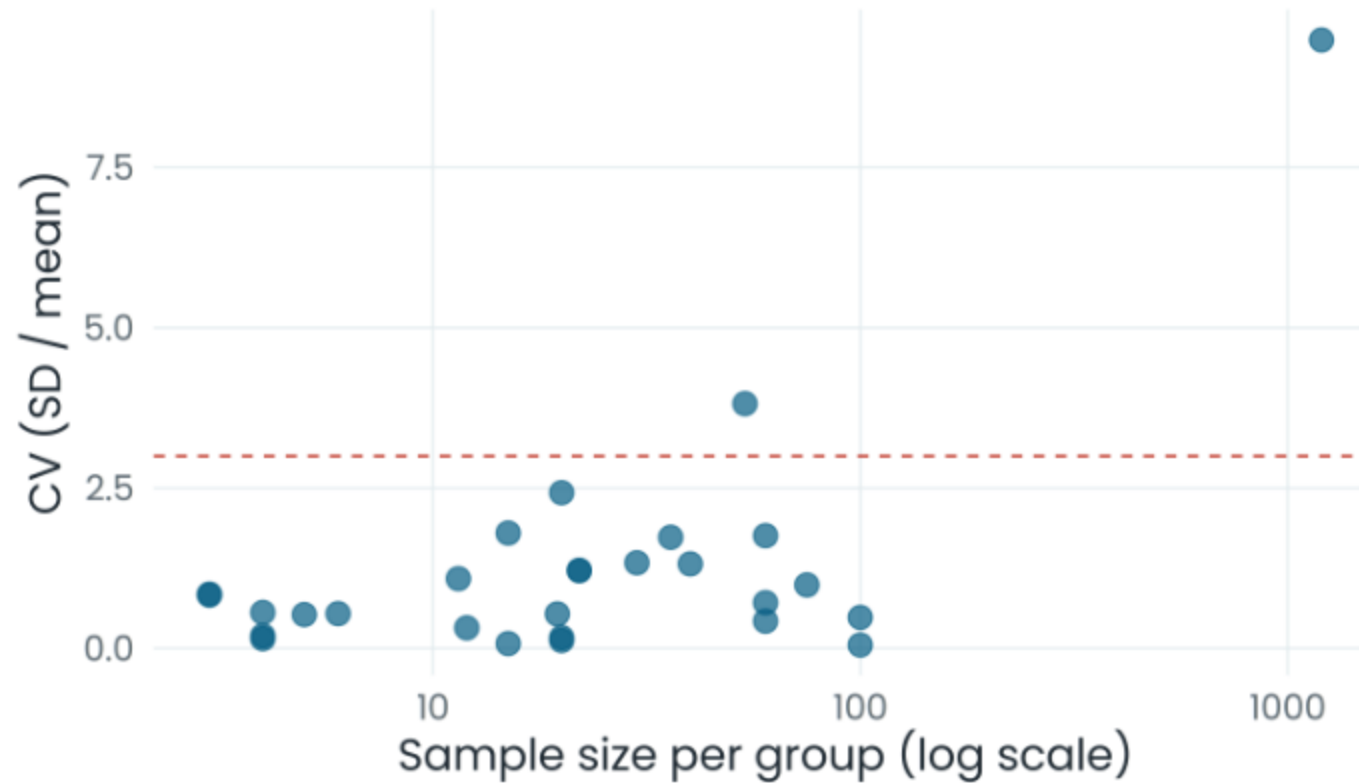
Rule	Why
Choose moderators a priori , in the protocol	Testing many moderators guarantees false positives
≈ 10 studies (not rows) per moderator level or coefficient	A between-study moderator is tested on the number of studies (remember $df \approx 10$ above): many rows per study add little
Report the omnibus test , coefficients and residual heterogeneity	“Significant” is not “explains the variation”
Beware confounding between moderators	Crops, regions and designs travel together
Prefer within-study contrasts when they exist	They control for study context
Moderator effects are associations , not causes	Studies were not randomised to moderator levels

Source: Thompson and Higgins (2002); Nakagawa et al. (2023b); Borenstein et al. (2021), chap. 20–21.

Missing data

Missing SDs: check before you impute

Pest studies with SDs, one point per study



To impute a missing SD for InRR, we borrow the **typical coefficient of variation** of the other studies (Nakagawa et al. 2023a).

First, look at the CVs: 2 studies stand out ($CV > 3$), one with $n = 1,200$ per group. Their SDs were **rebuilt as $SE \times \sqrt{n}$** : if n counts subsamples rather than true replicates, both the SD and the precision are wrong. **Check the paper.**

Missing SDs: impute, then compare

Pests (suspect studies set aside): 53 of 376 comparisons have no usable SD — 45 of them from **one** study.

Why the CV? The variance of $\ln RR$ is

$$v \approx \frac{SD_T^2}{n_T \bar{x}_T^2} + \frac{SD_C^2}{n_C \bar{x}_C^2} = \frac{CV_T^2}{n_T} + \frac{CV_C^2}{n_C}$$

So a missing SD only needs a **typical CV**, borrowed from the other studies:

$$v_{\text{imputed}} = \frac{\bar{CV}_T^2}{n_T} + \frac{\bar{CV}_C^2}{n_C}$$

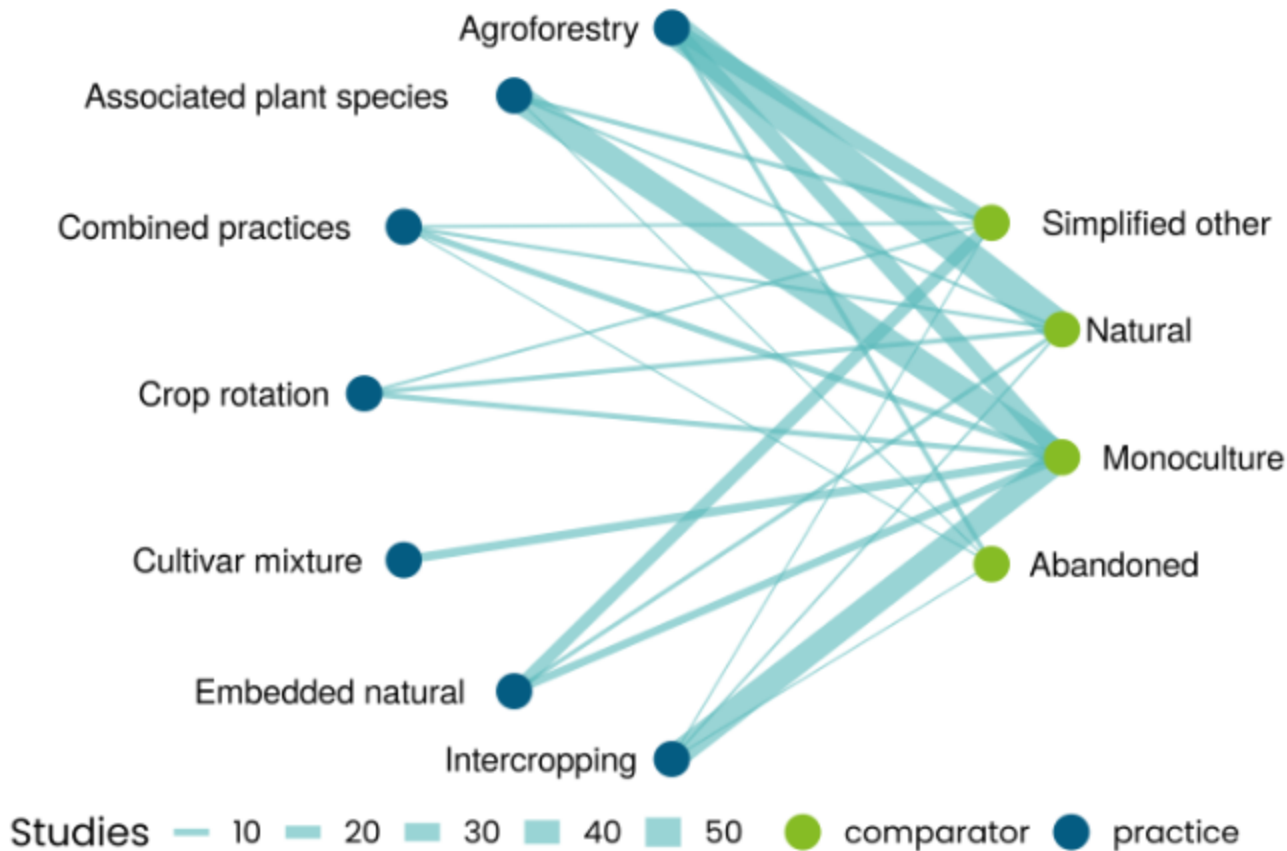
$\bar{CV}$: mean CV weighted by n , per group [Nakagawa et al. (2023a); metafor `vtype = "AV"`]. The screening matters: squared mean CV = 0.21 without the suspect studies, 28 with them.

	Pest abundance [95% CI]
Complete cases (323)	-28% [-42; -11]
With imputed SDs (376)	-30% [-45; -11]

Same conclusion — but most imputed rows come from one study, so do not over-generalise. Report **both**. Effect sizes: **recompute** them (from t , p , figures, medians), never fill them with averages.

Beyond one comparison

Network meta-analysis: comparing the never-compared



Intercropping or agroforestry: which is better? **Only 1 of 237 studies tested both.**

Network meta-analysis combines direct and **indirect** evidence (A vs C and B vs C → A vs B) to rank practices (Salanti 2012).

Here: a **star** around Monoculture, no closed loop → consistency cannot be checked; and **transitivity** is doubtful (the “monoculture” of agroforestry studies ≠ that of intercropping studies).
Tools: `netmeta`, `rma.mv`.

Other tools you may meet

Method	When	Keep in mind
Multivariate meta-analysis	Several correlated outcomes (e.g. biodiversity and yield on the same plots)	Needs the within-study correlation — test a range (van Houwelingen et al. 2002)
Selection models	Sensitivity to publication bias	Need many studies; ignore dependence
Random forests (MetaForest)	Exploring many candidate moderators	Exploratory only; weight by precision; cross-validate by study
Bayesian meta-analysis	Few studies, informative priors, complex structures	Priors must be justified and tested

Practical · run, read, discuss — in pairs

Time	Task	You...
16:33	Meta-regression: pests vs natural enemies	fill 1 blank , read F , coefficients, R^2
16:40	Within vs between studies	fill 1 blank , compare the two estimates
16:47	Missing SDs: CVs, impute, compare	run the given code, interpret
16:55	Debrief and wrap-up of the day (until 17:00)	

The script is **almost complete**: the work is **reading the outputs**, not writing code. Questions to discuss are in the script (`# Q:`). Open `meta_analysis_course.Rproj` (the course kit), then `td_advanced.R`; folded answers on the practical web page.

KEY MESSAGES

- 1 | Meta-regression tests whether a moderator **explains** heterogeneity; report the omnibus test, coefficients **and** what remains unexplained.
- 2 | Between-study moderators are **associations**; within-study contrasts are stronger.
- 3 | Missing SDs: **check CVs**, impute transparently, show results **with and without**.
- 4 | Advanced methods (network, multivariate, machine learning) answer specific questions — they do not rescue weak data.

In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- Moderators, missing data and networks
- Certainty of evidence and what can be inferred
- Your turn: soil carbon under agroforestry (end-to-end exercise)
- Before you submit: a checklist

literaturesynthesis.github.io/notebook

References

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